Sound Event Detection

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Abstract

The goal of automatic sound event detection (SED) methods is to recognize what is happening in an audio signal and when it is happening. In this experiment, I implement the CRNN model on a weak labeled dataset to recognize at what temporal instances different sounds are active within an audio signal.

In section1, I introduce the dataset, the framework of baseline model and the challenges in weak supervision In section 2,I implement the CRNN model shows some experiments about adjusting hyper parameters in detail. In section 3, I investigates ways of data augmentation.

1 Baseline model analysis

1.1 Dataset

The dataset provided by the task contained ten categories of 1579 + 3141 sub-events obtained from 10-second audio clips recorded or synthesized in a home environment to simulate a home environment.

1579 training events are weak labeled and 3141 evaluation events are strong labeled.

Strong label provides the source and event category within the audio as well as the start and end timestamps of the sound events.

Weak label only provides the categories of the sound events are provided without any timestamps.[1] Figure 1 shows how weak labels work.

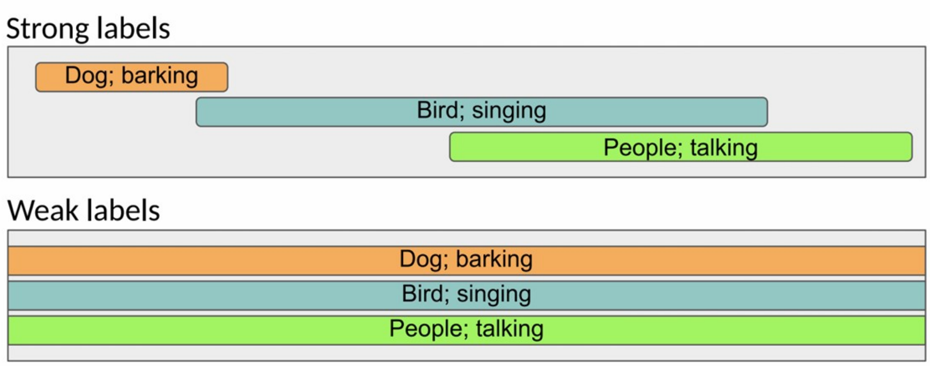

Figure 1: Weak label

In practice, we use log-Mel power spectrograms (LMS) as the feature of the audio clip with $D = 64$.

1.2 challenges

Our goal requires the identification of the sound source and event category for each audio segment, along with predicting the onset times of the sound events. While this weakly supervised task is more challenging than the strongly supervised task because we cannot directly extract information from the weakly labeled sound event segments. Therefore, alternative methods are necessary to identify the sound event segments.

1.3 baseline model

The pipeline of SED is shown in Figure: 2

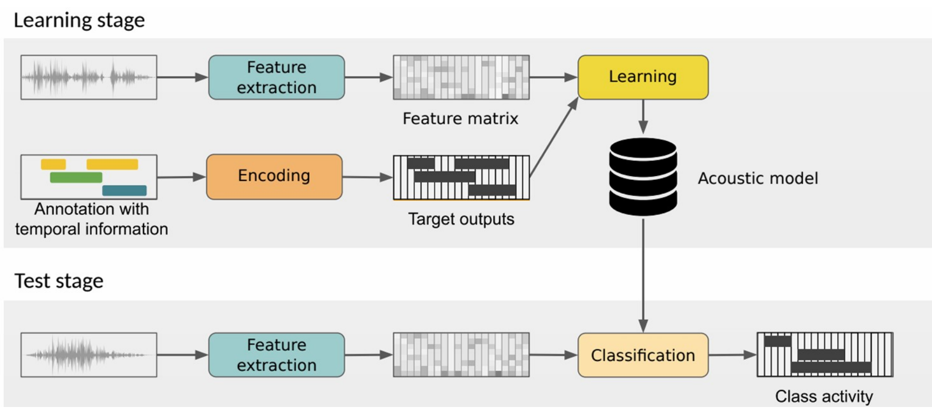

Figure 2: Pipeline

In data processing stage, the baseline uses a feature extractor(with a threshold to distinguish the sound event and background noise) to get a feature matrix that represents the input audio, and encodes the labels to produce the target outputs. Then, by weakly supervised learning, I trained a CRNN model[3] that can detect the category and on-offset of sound events in an audio clip.

The details of CRNN model is shown in the Figure:3. The input are pre-processed feature matrix $\mathbf{X} \in \mathbb{R}^{T \times D}$ which contains time-frequency information. The first layer in CRNN is a batch normalization. Then, there are three blocks, where each consists of a convolution layer, batch normalization, ReLU activation and maxpooling layer.

Next, for extracting temporal information, the data will be processed by a bidirectional GRU followed by a fully-connected layer. Moreover, to get a clip-level prediction, we exclusively utilize linear softmax and a Sigmoid layer to produce the outputs. This output represents the probability scores of sound events occurring over time. As this is a classification task, the loss function used during training is Binary Cross-Entropy (BCE) loss, defined as:

$$\text{Loss}(x, y) = -x \log(y) + (1 - x) \log(1 - y) \quad (1)$$

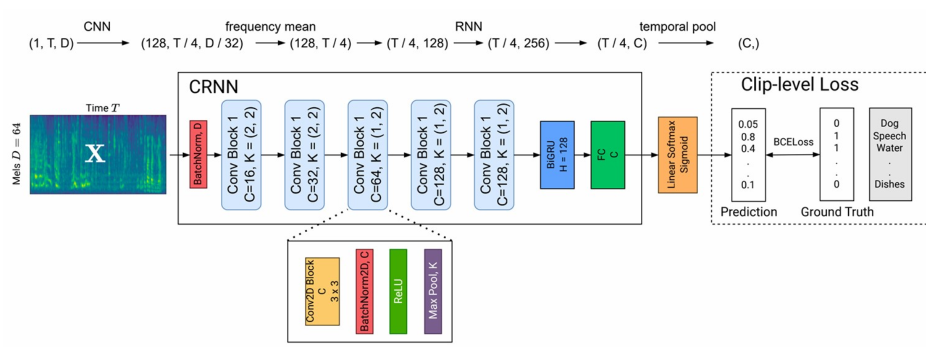

Figure 3: CRNN framework

2 Experiment

In this section, I do some experiments with different hyper-parameters and get my best model.

2.1 Baseline settings

The baseline model employs CRNN architecture with 5 Conv blocks and a 128D BiGRU with 2 layers. I use Adam optimizer with 0.001 learning rate.

For evaluation, we will introduce three types of metrics with F1 score, precision and recall rate as shown in Figure:6

1. Event based: measures on- and off-set overlap between prediction and ground truth and describes a model's capability to estimate a duration. See figure 4.
2. Segment based: measures of a given model's sound localization capability by the segment level overlap between ground truth and prediction. This can be seen as a coarse localization metric since precise timestamps are not required. See Figure 5
3. Tagging based: measures the models'capability to correctly identify the presence of an event within an audio clip. It shows the model's classification capability.

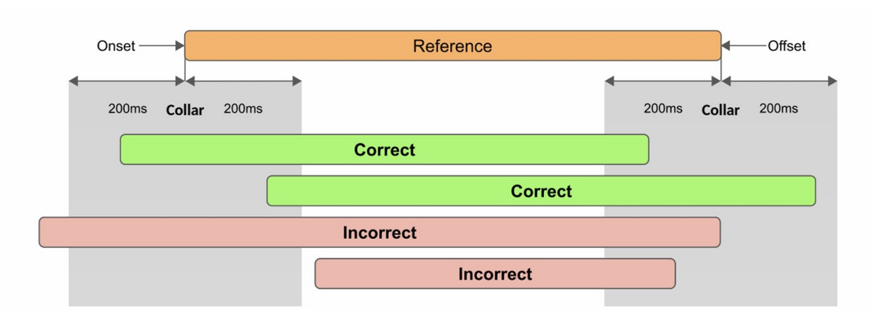

Figure 4: event F1 score

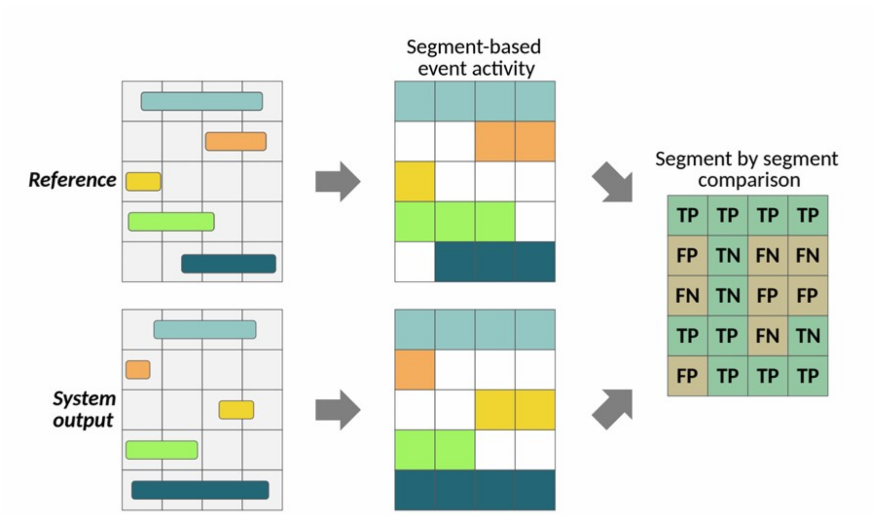

Figure 5: segment F1 score

2.2 Model optimization

2.2.1 Conv blocks

I changed the number of convolution blocks from 2 to 5, and the results are shown in Table 1.

3 conv blocks model has the best performance.

	f_measure	precision	recall
event_based	0.110395	0.0969593	0.165534
segment_based	0.593019	0.599425	0.605571
tagging_based	0.651152	0.664747	0.64995

Figure 6: baseline evaluation metric

model	mAP	Eve-F	Eve-P	Eve-R	Seg-F	Seg-P	Seg-R	Tag-F	Tag-P	Tag-R
2conv blocks	0.63	0.09	0.07	0.14	0.55	0.55	0.57	0.59	0.62	0.57
3conv blocks	**0.67**	**0.11**	**0.09**	**0.16**	**0.59**	**0.60**	**0.61**	**0.65**	**0.66**	**0.65**
4conv blocks	0.65	0.07	0.06	0.13	0.56	0.53	0.61	0.62	0.63	0.62
5conv blocks	0.65	0.01	0.02	0.01	0.24	0.43	0.0.17	0.61	0.62	0.62
2GRU layer	**0.67**	**0.11**	**0.09**	**0.16**	**0.59**	**0.60**	**0.61**	**0.65**	**0.66**	**0.65**
3GRU layer	0.64	0.07	0.06	0.13	0.56	0.53	0.61	0.62	0.63	0.62
4GRU layer	0.64	0.08	0.08	0.14	0.54	0.51	0.59	0.59	0.62	0.58

Table 1: model optimization
mAP represents the mean of the average precision in clip-level. "Eve-", "Seg-", "Tag-", "F", "P" and "R" mean event based, segment based, tagging based, F measure, precision, and recall, respectively. Baseline has 3 Blocks, 2 GRU layers.

2.2.2 GRU layers

As I kept the number of conv blocks on 3, I changed the number of GRU layers from 2 to 4, and the results are shown in Table 1

2 GRU layers model has the best performance.

3 Data augmentation

In this section, I investigate and survey methods of data augmentation[2] in audio learning, such as time stretching, pitch shifting, dynamic range compression, sub-frame time shifting, and block-mixing.

1. Noise Augmentation (NA): For every element in audio feature matrix $X \in \mathbb{R}^{T \times D}$, we add noise $x_n \sim \mathcal{N}(0, 1e-5)$ to it with probability of 0.5.

2. Time Shift Augmentation (TS): First, we choose a shift step $s \sim \text{int}(\mathcal{N}(0, 50))$. Then, we roll the feature matrix in the time domain.

3. Time Mask Augmentation (TM): First, we randomly choose an integer mask duration d from [0,60]. Then we randomly choose a start timestep ts and set feature value to zero within $[ts, ts+p]$. This operation will be executed twice.

4. Freq Shift Augmentation (FS): First, we choose a shift step $s \sim \text{int}(\mathcal{N}(0, 4))$. Then, we roll the feature matrix in the frequency domain.

5. Freq Mask Augmentation (FM): First, we randomly choose an integer mask duration d from [0,8]. Then we randomly choose a start timestep ts and set feature value to zero within $[ts, ts + p]$. This operation will be executed twice.

As I kept the number of conv blocks on 3 and the number pf GRU layers on 2, I tested the data augmentation method.

Experimental results with different data augmentations are shown in Table 2.

Data augmentation for frequency domain will contribute to higher scores in tag-based metrics for the improvement of the robustness of frequency information. In other words, data augmentation for frequency domain improves the ability of detecting.

And adding noise contributes in event-based metrics for the improvement of the robustness of temporal information. In other words, noise can help model develop the capability of localizing

DA	mAP	Eve-F	Eve-P	Eve-R	Seg-F	Seg-P	Seg-R	Tag-F	Tag-P	Tag-R
no DA	0.67	**0.11**	**0.09**	0.16	0.59	0.60	0.61	0.65	0.66	0.65
NA	0.67	0.11	0.09	**0.17**	0.59	0.60	0.60	**0.66**	0.67	0.66
TS	0.65	0.08	0.07	0.14	0.58	0.56	0.60	0.62	0.64	0.61
FS	**0.68**	0.08	0.07	0.15	0.58	0.57	0.61	0.64	0.64	0.64
FM	0.66	0.08	0.07	0.13	0.58	**0.59**	0.59	0.64	**0.67**	0.62
TM	0.63	0.08	0.06	0.16	**0.56**	0.55	**0.57**	0.61	0.64	**0.59**

Table 2: Data augmentation
”NA”, ” TS”, ”FS”, ”FM”, ”TM” means noise augmentation, time shift, frequency shift, time mask, frequency mask.
mAP represents the mean of the average precision in clip-level. ”Eve-”, ”Seg-”, ”Tag-”, ”F”, ”P” and ”R” mean event based, segment based, tagging based, F measure, precision, and recall, respectively.
Baseline has 3 Blocks, 2 GRU layers.

Conv blocks	GRU layer	Data augmentation
3	2	NA

Table 3: Best model

4 Conclusion

In summary, I have learnt the task of SED with its general model architecture. Plus, I have tried some experiments about adjusting hyper parameters, such as the number of blocks and the number of layers of the feature vector. Moreover, I investigate the data augmentation and explore their influences. I uses 3 types of metrics measure the performance. Considering Event-based F1 score, I list the best result and the parameters of it in Figure 6 and Table 3

References

[1] Heinrich Dinkel, Mengyue Wu, and Kai Yu. Towards duration robust weakly supervised sound event detection. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 29:887–900, 2021.

[2] Annamaria Mesaros, Toni Heittola, Tuomas Virtanen, and Mark D. Plumbley. Sound event detection: A tutorial. *IEEE Signal Processing Magazine*, 38(5):67–83, September 2021.

[3] Baoguang Shi, Xiang Bai, and Cong Yao. An end-to-end trainable neural network for image-based sequence recognition and its application to scene text recognition. 2015.